

Turner M. Bumbarly

Home: 703-860-3654 · Cell: 202-864-9623 · turnerbumbarly@gmail.com

Education

M.S. Aerospace Engineering | Georgia Institute of Technology – Atlanta, GA

Dec 2025

B.S. Mechanical Engineering | California Institute of Technology – Pasadena, CA

Jun 2024

Technical Skills

Programming Languages: Matlab | Python | ROS2 | LabVIEW | Java | C/C++ | Objective C

Engineering Skills: SolidWorks | FEA | GD&T | Rapid Prototyping | Machining | Soldering / Elec.

Professional Experience

2024 – Present

Aerial Robotics and Experimental Autonomy Lab | Graduate Student

Project lead for autonomous drone demonstration focused on anti-UAV mission to detect and capture enemy drones; Developed simulations to model anti-aerial capabilities in virtual testbed environment; Directed real-world flight tests, executing autonomous flight control on Holybro X500 drone to validate system performance

2023 – 2024

NASA Jet Propulsion Laboratory | Mechanical Engineering Intern

Led mechanical design of long-range, autonomous drone for LORNA Martian surface scouting mission; Designed mechanical apparatus to measure the thermal conductivity of lunar soil; Created LabVIEW test script to characterize the performance of deformable mirrors in the IRIS telescope testbed

2022 – 2022

National Reconnaissance Office | Systems Engineering Intern

Supported government acquisition of space systems for national security; Briefed general officers/senior executives; Collaborated with stakeholders across the IC

2021 – 2024

Founder & Project Manager | Caltech CubeSat Club

Raised >\$10,000 in seed funding via research grants; Expanded team from 3 to 20+ students; Organized product development reviews with NASA scientists/professionals

2020 – 2021

Google Computer Science Summer Institute

Developed weekly projects with students and professionals; Learned JavaScript and HTML from Google Developers; Conducted live code-alongs with Google Engineers.

2020 – 2020

Research Coordinator | Texas State University

Managed over a dozen student research projects; Conducted applied math research on “Multicore Computer Algorithms” with Dr. Rodion Podorozhny

Leadership & Involvements

2024 - Present

Graduate Student Senate

Managed \$1.2 million student activities budget for campus orgs and events; Passed legislation to reform student-advisor relationship; Representative for school of aerospace engineering

2020 – 2024

Air Force ROTC

Type 1 scholarship recipient (covers full tuition and fees); Awarded Daedalian Scholarship (\$3,000) for exceptional performance; Honorary Flight Award